



Eleventh International Conference on Biosignals, Images, and Instrumentation (ICBSII-2025)

Day 1 - Track 1: AI & DS for Healthcare

Venue: CHT - I Floor, BME, 02:30 p.m. – 04:00 p.m. IST

Internal Session Chair: Dr.S.Saranya
Student POC: Mr.S Vidhya Sagar

Sl. No.	Paper ID	Paper Title	Authors
1	48	IoT Based Gesture-Controlled Mouse System for Individuals with Hand Impairments	Aldarin Jino , Deva Priya Isravel, Julia Punitha Malar Dhas
2	101	Prediction of Epileptic Seizures Using Liquid Time-Constant Networks	Darshan Shestha , Saranya G, Padmanabha Sarma, T. Swethaa, Manjushri M
3	215	Detecting Schizophrenia through EEG Signals using Deep Learning	Ashish Make , Swarnalata Bollavarapu, Radhika Baid, Devansh Gupta
4	86	MINDCHRONO: A Comparative Study of Deep Learning Models for EEG-Based Depression Detection	Balakrishna N , M.B. Mukesh Krishnan, D. Ganesh
5	124	Adaptive Deep Knowledge Framework for Classifying Sleep Stage using Deep Feature Learning	Amoga Lekshmi S , Deepthi R, Amutha R
6	366	A Framework for Early Detection and Analysis of Prenatal Brain Development	Sathiyavathi S , G. Udhayakumar
7	243	A Medical Chatbot for Diabetic Patients Using Artificial Intelligence	Mangai S ,A.Supriya,V.Kavitha
65	117	Enhanced MRI-Based Brain Tumor Detection Using Deep Learning Architectures and Optimized Machine Learning Techniques	Narenthirakumar Appavu



Eleventh International Conference on Biosignals, Images, and Instrumentation (ICBSII-2025)

Day 1 - Track 2: Biomedical Imaging and Image Processing

Venue: BMI - Ground Floor, BME, 02:30 p.m. – 04:00 p.m. IST

Internal Session Chair: Dr. K. Nirmala
Student POC: Ms. Harishini H

Sl. No.	Paper ID	Paper Title	Authors
8	123	Diabetic Retinopathy From Retinal Imaging	Soorya, Narayanan V, Mohammed Anfaal D, Rajadurai S, Akash V, Divya Bharathi S, Sandhya C
9	79	Particle Swarm Optimization of CNN for Early Parkinson's Disease Detection	Sharvesh T, Srinivasan, Shrivanshi Ganesh Pai, Basavaraj Hiremath, Ramsha Fathima, Parthib Basu
10	52	Analysis of Blood flow in Carotid Artery at Various Blood Viscosities During Rest and Physical Exercise	Sankar Kandasamy, Kalpana, Ramakrishnan, Malathi M
11	370	Deep Learning-Based Automated Breast Cancer Ultrasound Image Classification: A Study	R Sivakumar
12	157	Transfer Learning with Deep Convolutional Neural Networks for Esophageal Cancer	Susendhiran M, Akshaya A, Jeevitha S, Gowri S
13	158	Hybrid Ensemble Methods for Acute Lymphoblastic Leukemia Diagnosis: Integrating CNN Features with Artificial Rabbits Optimization Algorithm	Divyadharshini K, Suvetha R, Alagu S, Harithra R
14	374	Image Enhancement using linear filters for Rectal Cancer Diagnosis using MR Images	Sonal patreena D, Almeida, Surekha Kamath, Rajgopal KV, Manjunath KN

66	119	Deep Learning for Predicting Invasive Ductal Carcinoma in Histopathological Tissue Images	Narenthirakumar Appavu
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Eleventh International Conference on Biosignals, Images, and Instrumentation (ICBSII-2025)

Day 1 - Track 3: Biomedical Sensors, Instrumentation and Wearable Systems

Venue: DTE Lab II Floor, BME, 02:30 p.m. – 04:00 p.m. IST

Internal Session Chair: Dr.Suganthi L
Student POC: Ms.B.Madhumitha

Sl. No.	Paper ID	Paper Title	Authors
15	361	Thermogravimetric Analysis and Tensile Testing of PEEK Structures Fabricated via Extrusion and 3D Printing	Marieswaran M , Ravi Tiwari, Somnath Bhattacharya, Sanjeev Kumar
16	244	Software for Clinical Speech Therapy Services	Mangai S, S.Govindaraj,K.Anjana
17	352	A Smart Shoe for Detection and Alleviation of Foot Ulcers	Sri Balaji P , Vishnu N, Sai Mathangi S, Dr. S. Poonguzhali
18	126	A Study of Spectral Preprocessing Techniques for Bilirubin Quantification Using DRS	Chitra R , Shanthi Prince
19	149	Self-Stabilized Gyro Toothbrush For Parkinson's Disease: Enhancing Oral Hygiene Through Real-Time Motion Compensation	B. Nivethitha , A. Kirubhashini, A. Josephin Arockia Dhivya, Jaya Rubi
20	142	Terahertz-Based Biosensor for Characterization of Cancerous Cells	Hemalatha E , Lalit Dhakshan M, Karthick R, Meraline Selvaraj, Dr. Sreeja B S
21	164	OptiVue - A Smart Approach to Eye Health Screening	Swetha A , Kiruthikha M, Ramyaa G R, Dr. S. Prabhakar, Sivagowrika S
78	121	Improved Retinopathy Detection Using ResNet-50 Optimized AI Deep Learning Models	Narenthira Kumar Appavu



Eleventh International Conference on Biosignals, Images, and Instrumentation (ICBSII-2025)

Day 1 - Track 4: Biomedical Signal Processing

Venue: MSDM Lab I Floor, BME, 02:30 p.m. – 04:00 p.m. IST

Internal Session Chair: Dr.B.Geethanjali
Student POC: Ms.Nivithasri Selvaraj

Sl. No.	Paper ID	Paper Title	Authors
22	104	VLSI Architecture for Modulating Nerve Activity in Parkinson's Disease Using STFT Transformation Methods	S. Karthick , Neelamegam Devarasu
23	19	AI Powered Heart Rate Monitoring System Leveraging Skintone	Kanimozhi V , V. P. Kalaiyarasi, Danish Mithula S, Kishore I M
24	319	Waves Don't Lie: Leveraging Test-Time Training and Kolmogorov Arnold Networks for EEG-Based Biometrics	Eshwanth Karti T R , Nithish Ariyha K, Vikash J, Yeshwanth Balaji AP, Amrutha Veluppal
25	118	Evaluation Methods for Studying Muscle Fatigue through Physiological Signal Monitoring	Annapoorani CL , Sofia Bobby J, Subha Ramya, Miruthula A, Mythily V
26	248	Preterm Birth Prediction Using GraphSAGE and Electrohysterogram Data	Nandhini Kuppala , Monish GV, Nivedha V, Balu Pinnenti, Amrutha V
27	278	ESP32: Smart Support Empowering Alzheimer's Patients	Aloysious Alex , Alen Christy C R, Adityan Ramesh, Hridhya K H, Anju Babu, Siji Joseph
28	176	MaskCycleGAN-based Whisper to Normal Speech Conversion	Johanan Joysingh S , Rohith Gupta K, Ramnath K, Vijayalakshmi P, Nagarajan T



Eleventh International Conference on Biosignals, Images, and Instrumentation (ICBSII-2025)

Day 1 - Track 5: Materials for Bioscience

Venue: BME Seminar Hall III Floor, BME, 02:30 p.m. – 04:00 p.m. IST

Internal Session Chair: Dr.R.Subashini
Student POC: Ms.Shekinah A

Sl. No.	Paper ID	Paper Title	Presenter Name
29	156	Predictive Analytics in Dentistry: AI Applications in Frenectomy and Soft Tissue Management (Online)	Ashna Yuvaraj , Pradeep Kumar Yadalam, Karthickraj S, M. Soundharya Manogaran, Swarnambiga
30	228	Segmentation Of Neutrophil Nucleus in White Blood Cell Images Using Kapur's Entropy-Based Improved Grey Wolf Optimization	Saranya R , Bakiya A
31	154	Machine Learning Prediction of Marginal Gingival Adaptation of Direct Composite Veneering with or without Mock-Up in Upper Anterior (154)	Ashna Yuvaraj , Swarnambiga Ayyachamy, Pradeep Kumar Yadalam, Soundharya Manogaran, Karthickraj S.
32	78	Indian Herbal Formulation for Repigmentation of Vitiligo	K. Mythili , S. Prabakar
33	42	Incorporation of Banana Leaf Extract into Collagen Sheets for Fire Burn Healing	Brindha M , Nareshkumar N, Prabakar S, Nathish B, Dharanya S, B. Poovarasan
34	321	Design of Components for a 3D Printed Assistive Robotic Arm Optimized for Precision Surgical Procedures	Jaya Rubi , T. R. Thamizhvani, R. J. Hemalatha, B. Varshini, A. Josephine Arockia Dhivya, S. G. Viena



Eleventh International Conference on Biosignals, Images, and Instrumentation (ICBSII-2025)

Day 2 - Track 1: AI & DS for Health Care

Venue: CHT-I Floor, BME, 02:30 p.m. – 04:00 p.m. IST

Internal Session Chair: Dr.S.Saranya
Student POC: Ms.Shrinithi SK

Sl. No.	Paper ID	Paper Title	Presenter Name
35	87	A Novel & Hybrid Algorithm for Real-Time Depression Detection via EEG Analysis	N. Balakrishna , M. B. Mukesh Krishnan, D. Ganesh
36	181	Wheat Disease Detection: Bridging the Gap with Deep Learning Approaches	Vijaykumar Vasaiya , Amit Vajpayee, Ankita Gandhi
37	165	Improving Hepatitis-C Detection Accuracy Using Improved C5.0 Algorithm with Chi-Square Feature Selection	Shamitha G , S. Alagu, Sushmitha A, Shohail Shahabudeen
38	178	Southern Regional Healthcare Chatbot Using AI and NLP	S. Raja Mohamad , Muthukamalesh V, Sreeju S. A, Akaash Henry V
39	138	AI-Based Decision Support System for Body Constitution (Yakkai Ilakkanam) Classification Using Minimal Optimized Textual Features	Keerthivasan G , Sutha S, Santhosh K M, Pappa N, Praveen Kumar L
40	258	A Study On Cardiovascular Disease Detection In Diabetic Patients	G. Anitha
41	344	Chronic Kidney Disease Prediction Using Deep Learning	Hepsiba D , Vignesh R



Eleventh International Conference on Biosignals, Images, and Instrumentation (ICBSII-2025)

Day 2 - Track 2: Biomedical Imaging and Image Processing

Venue: BMI - Ground Floor, BME, 02:30 p.m. – 04:00 p.m. IST

Internal Session Chair: Dr.R.Nithya
Student POC: Mr.Sujay

Sl. No.	Paper ID	Paper Title	Presenter Name
42	281	Automated Deep Learning System for Early Detection of Stroke Using CNN Architecture	Kshnika O , S. Gunasekar, G. Babu, M. Neela Harish
43	332	Computational Linear Modeling for Human Lung Mechanics	L. Priya , Vishnupriya B, Dhanusree N, Yogeshwaran S, Priyadharshni S
44	72	Prosthetic Hand Grasp Recognition Using Residual-Unet Combined with Cross Over Attention	Premalatha G , Srikanth Vasamsetti, Srivatsal Narayan Sekar, Anup Chander
45	373	Early Detection of Liver Disease and Precise Health Optimization Using Advanced Computational Models	Rubin Bose S , Angelin Jeba J, S. Viswanathan, V. Mahalakshmi, V. Hari, Regin R
46	153	Detection of Finish Lines in Indirect Ceramic and Composite Veneers Using Clinical Images with the Help of Developed AI-Based System (Online)	Ashna Yuvaraj , Soundharya Manogaran, Pradeep Kumar Yadalam, Swarnambiga Ayyachamy, Karthick Raj
47	50	Automated Pneumonia Identification Using Convolutional Neural Network	Gnanapriya M , V. Loganathan, S. Prabakar, Revathi N, Rakshagamoorthy V, Karthiswaran M
48	76	Multimodal Analysis of Hereditary Breast and Ovarian Cancer Syndrome Using Deep Learning Techniques	Dhivya Pradeep , Subham Lakshmi B, Shanthini Devi P, Kavitha G, Thaarini Devi S



Eleventh International Conference on Biosignals, Images, and Instrumentation (ICBSII-2025)

Day 2 - Track 3: Biomedical Sensors, Instrumentation and Wearable Systems

Venue: DTE Lab II - Floor, BME, 02:30 p.m. – 04:00 p.m. IST

Internal Session Chair: Dr.Subitcha J
Student POC: Ms.Ezhil Meenaa A S

Sl. No.	Paper ID	Paper Title	Author Name
49	325	Ergonomically Optimized Wearable Seating Posture Monitoring and Alert System for Mitigating Work-Related Musculoskeletal Disorders	Christopher Dennis J , Raksana B, Rakshita T S, Dr. Poonghuzhali S
50	166	Flexural Transducer Based Monolithic Diathermy Probe for Thermo Therapy	Nithyakalyani K , Nanthini B, N. Bharathiraja, Vishnupriya A, Madhumitha K, Harinni K K
51	167	CARE-DAQ: Low Cost Open Source Wireless Biomedical Data Acquisition System	Ramakrishnan Maharajan
52	312	Real-Time Health Monitoring and Tracking System	Shelma Vincent V. S , Hepsiba D, Amala Hepzibha D, L. D. Vijay Anand
53	120	Self Powered Energy Harvesting Wearable Monitor	Sofia Bobby J , Mythily V, Annapoorani C. L, Reena Shree, Subha
54	323	Smart ICU Revolutionizing Critical Care with Multi-Modal Sensing and MIoT-Driven Solutions	Iyyanarappan A , Shakira Banu S, Madhumitha B.
55	111	TheraLink Smart Health Multiparametric Monitoring System with Wireless Data Transmission	Rohan Steven B , Anne Joan Benita V, Soorya Narayanan V, Sandhya C, S. Vinurajkumar, Divya Bharathi S



Eleventh International Conference on Biosignals, Images, and Instrumentation (ICBSII-2025)

Day 2 - Track 4: Biomedical Signal Processing

Venue: MSDM Lab I Floor, BME, 02:30 p.m. – 04:00 p.m. IST

Internal Session Chair: Dr.M.Dhanalakshmi
Student POC: Ms.Soundharya

Sl. No.	Paper ID	Paper Title	Author Name
56	233	Machine Learning Based Categorization of Heart Sounds for Cardiac Arrhythmia Detection	Sreedevi Nair , Indu T
57	330	Biophysical and Anthropometric Determinants of Muscle Co-activation Patterns during Human Gait	Arunthathi Sivalingam , Saranya S
58	303	HealthNet: Secure and Scalable Hospital Network for Enhanced Patient Care	M. Vivek Kumar , K. Sureka, S. Shakthi Poovannan, M. Thirisankari, V. Soundharya
59	324	Using Web Mining to Track and Predict the Spread of Diseases	Sadik Khan
60	342	Single-Modality Emotion Detection: EEG-Based Feature Engineering and Interpretability	Vishal A S , Deepan S, Amrutha V
61	322	Realtime Cardiac Health Monitoring Using AI-Powered Embedded Systems	Tarun V. M , Madhumithaa S, Masoodhu Banu N. M, Jehosheba Margaret M
62	43	Leveraging Spectral Signatures and Attention Enhanced BiLSTM Network for Comprehensive Respiratory Sound Analysis	Rohit VP , M V V S Swaroop, Mithin Dev A, Suriya KP, Amrutha Veluppal



Eleventh International Conference on Biosignals, Images, and Instrumentation (ICBSII-2025)

Day 3 - Track 1: AI & DS for Health Care

Venue: CHT-I Floor, BME, 11:15 p.m. – 12:30 p.m. IST

Internal Session Chair: Dr.J.Vijay
Student POC: Ms.Gayathri

Sl. No.	Paper ID	Paper Title	Presenter Name
63	175	Detecting Childhood Autistic Spectrum Disorder and Enhancing Their Reading Skills	Bino J , Pratheeksha S, Shameera S I.
64	180	Machine Learning in Mental Health: Bridging Gaps in Diagnosis and Intervention (online)	Nirav Kansara , Brijesh Vala, Mohammad Shahnawaz Shaikh
67	193	Voice Assistance for Visually Impaired Persons Using AI and IoT (online)	Ramesh L , Pruthiga M S, Muthulakshmi M, Puvitha Sri N, Navaneetha S
68	174	An Intelligent System for Heart Sound Signal Analysis: A Survey	Manu Saraswat , Chhote Lal Prasad Gupta
69	155	Evolution Scale Modelling for Prediction of Regulator G Protein Signalling (RGS) Sequences in Oral Cancer	Soundharya Manoharan , Swarnambiga Ayyachamy, Ramya Ramadoss, Pradeep Kumar Yadalam



Eleventh International Conference on Biosignals, Images, and Instrumentation (ICBSII-2025)

Day 3 - Track 2: Biomedical Imaging and Image Processing

Venue: BMI Ground Floor, BME, 11:15 p.m. – 12:30 p.m. IST

Internal Session Chair: Dr.S.Suresh Anand
Student POC: Ms.Nithiyaa Dharshini

Sl. No.	Paper ID	Paper Title	Presenter Name
70	314	Myoelectric Controlled Lower Limb Prosthesis with Active Ankle-Foot Joint	M. Neela Harish , Harini Sri Kamatchi K, S. Sathish, S. Prasanthi, Anushiya K
71	349	An Advanced Smart Glove for Stroke Rehabilitation Using IoT	Dharun R
72	274	Simulation and Experimental Study of Quartz-Enhanced Photoacoustic Spectroscopy Technique for Human Exhale Breath Gas Sensing	Saran Krishnamoorthy , Saran Kumar, Kishore S, Sooraj Shiby, Nilesh J Vasa
73	313	Injectable Hydrogel Loaded with Copper Oxide–Curcumin Nanosensitizer for Enhanced Antibacterial Sonodynamic Effect	Abiramy Soundararajan , Preethi Vasundara N, Bhavya V, Akshaya A. P.
74	225	A framework of a smart therapeutic insole system by measuring plantar pressure for early diagnosis of diabetic foot ulcers using machine learning and thermal imaging techniques	Diviya S , Sangamitra N, Venuhasini S M, B. Suresh Chander Kapali, A. Saravan
75	182	A Deep Learning Approach to Multi-Class Skin Disease Identification	Hariharan R , Dhanushraju S, P. Dharshini Muthu Bala, Saaral S, Devi Vijayan



Eleventh International Conference on Biosignals, Images, and Instrumentation (ICBSII-2025)

Day 3 - Track 3: Biomedical Sensors, AI, Instrumentation and Wearable Systems

Venue: MSDM Lab, I -Floor, BME, 11:15 p.m. – 12:30 p.m. IST

Internal Session Chair: Dr.S.Arun Karthick
Student POC: Ms.Aruna Devi

Sl. No.	Paper ID	Paper Title	Presenter Name
76	195	Hybrid Deep Learning for Detecting Fake Medical Images and Heart Sounds	Soundarya , R. Vijayalakshmi, P.Priya Dharshini
77	230	Intelligent Intravenous Drip Stand for Safe Infusions Using IoT Integration	Sushma Philipraj , Remya Raj
79	249	Hybrid Deep Learning Models for Early Detection of Chronic Kidney Disease: A1-D CNN and ResNet 152-V2 Approach	Jeena Jose , E. J. Thomson Fredrik
80	256	Harnessing Deep Learning for Thyroid Disease Detection: A Systematic Review	Fasira T , S. Jerine
81	341	EEG-Driven Massage Headband for Anxiety Alleviation (Online)	K. Sravani , Dr. Swathi Nadipineni, B. Anjali, Ch Vedavathi, B. Ramya
82	336	Histopathological Image Analysis Using Improved Gorilla Troops Optimizer for Lung Cancer Detection	Bharanidharan N , Nepolian V

Online Meeting Link - ICBSII 2025

<https://meet.google.com/cqf-jzue-eyh>